Smart Energy Consumption Visualization Dashboard Using Power BI

Literature Summary of Six Relevant Papers

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Abstract

This document reviews six studies related to smart energy monitoring, electricity-consumption analytics, interactive visualization, forecasting, and Power BI dashboards. The literature indicates that effective dashboards depend on reliable data collection, data cleaning, dimensional modeling, meaningful visual encoding, interactivity, and actionable analytics. These findings support the proposed project: a Power BI dashboard that helps users understand consumption trends, detect high-use periods, compare consumers or facilities, and support energy-efficiency decisions.

Review Scope

The selected papers represent both direct Power BI applications and complementary smart-energy research. Together they cover business-intelligence workflows, big-data visualization, IoT sensing, machine-learning prediction, and advanced visual analytics.

No.	Paper / Study	Main contribution
1	Energy Consumption Distribution in Portugal: A Business Intelligence Solution	Develops a BI conceptual model and interactive Power BI dashboard using monthly, hourly, and geographic consumption data.
2	Electricity Consumption using Big Data Analytics (Awate et al., 2023)	Analyzes 500 residential consumers over 24 months; combines Power BI visualization with Prophet forecasting and consumer ranking.
3	Energy Calculation Using Power BI (2023)	Describes an end-to-end dashboard workflow: collection, transformation, modeling, visuals, filters, testing, and sharing.
4	Visualization as a Means of Big Data Management: Qatar Electricity Consumption Data (2017)	Highlights the importance of standardization, data quality, speed, outlier handling, and visualization for energy-sector big data.
5	An IoT-Enabled Smart Energy Monitoring and Fault Prediction System	Combines ESP32 sensors, MQTT cloud communication, real-time dashboards, regression forecasting, and fault anticipation.
6	Interactive Machine Learning-Powered Dashboard (ESANN 2024)	Uses machine learning and interactive visual analytics, including t-SNE maps, data-cube filtering, sparklines, and morphing projections.

1. Energy Consumption Distribution in Portugal: A Business Intelligence Solution

This dissertation applies Design Science Research Methodology to build a BI solution for analyzing monthly and hourly electricity consumption across geographic areas. Its Power BI dashboard supports interactive exploration and decision-making through multiple datasets, making it directly relevant to the proposed project. The key lesson is to begin with explicit business questions and connect the data model, visuals, evaluation, and user feedback.

2. Electricity Consumption using Big Data Analytics

Awate and colleagues study 500 residential consumers across 24 months using electricity data and Power BI. The system identifies top consumers, average consumption by user, yearly trends, and high-consumption months; it also uses the Prophet model to forecast future monthly demand. The limitation is that monthly data restricts temporal granularity, suggesting that the proposed dashboard should support hourly or daily data where available.

3. Energy Calculation Using Power BI

This study presents a practical implementation sequence for an energy dashboard: collect data from smart meters, sensors, or bills; clean and transform it; build relationships and calculations; create charts and maps; add slicers; test accuracy; and publish for collaboration. It emphasizes usability and stakeholder-oriented layout rather than analytics alone. The proposed project can adapt this workflow using a star schema, DAX measures, KPI cards, trend charts, decomposition views, and drill-through pages.

4. Visualization as a Means of Big Data Management: Qatar Electricity Consumption Data

The Qatar-focused study frames visualization as a way to convert large and inconsistent energy datasets into useful information. It identifies challenges such as lack of standardization, data integration, speed, outliers, and difficulty displaying meaningful results. This paper is valuable for the data-engineering layer of the proposed project: validation rules, consistent units, missing-value treatment, and transparent refresh processes are essential.

5. An IoT-Enabled Smart Energy Monitoring and Fault Prediction System

This paper combines ESP32-based voltage and current measurement, calibration, MQTT transmission, cloud processing, real-time visualization, machine-learning regression, and fault prediction. Although its dashboard is web-based rather than Power BI-based, the architecture demonstrates how live sensor data can feed monitoring and prediction. For the proposed dashboard, IoT data can be stored in a gateway or cloud database and then connected to Power BI through scheduled or streaming mechanisms.

6. Interactive Machine Learning-Powered Dashboard

The ESANN study integrates machine learning with interactive visual analytics for residential energy-demand monitoring. Its contributions include two-dimensional maps of daily demand profiles using t-SNE, data-cube filtering, sparklines, and morphing projections. This work suggests a future enhancement for the proposed dashboard: clustering users or buildings by load shape and presenting the groups through interpretable interactive visuals.

Cross-Paper Findings

Across the six studies, five design requirements recur: reliable and standardized data; a clear ETL pipeline; a well-structured analytical model; interactive visuals that answer stakeholder questions; and predictive or diagnostic features that turn observation into action. Power BI is particularly suitable for the reporting and exploration layer, while Python, SQL, IoT platforms, or machine-learning services can provide data preparation and advanced analytics.

Proposed Dashboard Features

The proposed Smart Energy Consumption Visualization Dashboard should include: total energy and cost KPIs; daily, weekly, monthly, and hourly trends; peak-demand identification; consumer/building comparisons; top-N high-consumption entities; load-profile heatmaps; anomaly flags; renewable-versus-grid contribution where applicable; forecast charts; slicers for date, location, consumer, tariff, and appliance; and drill-through pages for detailed investigation.

Research Gap and Proposed Contribution

Existing work often focuses on either Power BI reporting, IoT monitoring, or machine-learning analytics. A useful contribution is to integrate these layers into one practical dashboard with documented data quality checks, role-specific views, explainable KPIs, and evaluation based on dashboard usability and decision usefulness. The project can measure success through refresh reliability, visual clarity, forecast error where forecasting is included, and the ability to identify actionable energy-saving opportunities.

References

1. Energy Consumption Distribution in Portugal: A Business Intelligence Solution, 2024 dissertation, NOVA IMS repository.
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